PIMamba: Physics-Informed Mamba for Few-Shot Hyperspectral Image Classification

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***Abstract*—Hyperspectral image (HSI) classification plays a crucial role in fields such as remote sensing, agriculture, and environmental monitoring. However, few-shot HSI classification remains a significant challenge. In this article, we propose PIMamba, a physics-informed HSI classification framework that integrates diffusion-inspired physical modeling into the state-space-based Mamba architecture. The proposed method models hidden-state evolution with physically guided spectral–spatial dynamics and introduces spectral physical loss (SpeLoss) and spatial physical loss (SpaLoss) to enforce physically consistent feature representations during training. In addition, a dynamic parameter adjustment (DPA) mechanism is developed to adaptively balance spectral and spatial physical constraints. Extensive experiments on four benchmark datasets [Indian Pines (IP), University of Pavia (UP), Salinas Scene (SA), and Houston 2013 (Houston)] demonstrate that PIMamba achieves state-of-the-art (SOTA) classification performance under few-shot settings without relying on external annotated data or large-scale pretrained models.**

***Index Terms*—Few-shot learning (FSL), hyperspectral image (HSI) classification, Mamba, physical constraints, physics-informed neural networks (PINNs).**

I. Introduction

HYPERSPECTRAL images (HSIs) capture information across hundreds of contiguous spectral bands, providing significantly more information than traditional RGB images. To transform the spectral information into actionable knowledge for practical applications, HSI classification becomes a critical step. Through HSI classification, ground objects can be accurately identified and categorized, enabling a wide range of remote sensing applications such as agricultural monitoring, forest fire detection, and environmental assessment.

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For HSI classification tasks, a variety of traditional machine learning approaches and deep learning paradigms have been explored. Traditional methods, such as support vector machines (SVMs) [1], random forests (RFs) [2], and k-nearest neighbors (KNNs), rely on feature selection and dimensionality reduction techniques [3]. Although these approaches can achieve stable performance on relatively small datasets, they have limited capability in modeling complex nonlinear relationships and high-dimensional spectral–spatial features. Deep learning methods, on the other hand, leverage complex network architectures such as convolutional neural networks (CNNs) [4], [5], graph neural networks (GNNs) [6], recurrent neural networks (RNNs) [7], and Transformers [8], [9], [10] to automatically extract high-dimensional features, significantly improving classification accuracy.

Recently, state-space-model (SSM)-based architectures have attracted increasing attention due to their capability in efficient sequence modeling. In particular, the emerging backbone network Mamba [12], based on SSMs [13], has been preliminarily applied in HSI classification, demonstrating potential due to its superior computational efficiency. Nevertheless, the performance of these deep learning methods generally relies on large-scale labeled datasets, and their high model capacity often leads to overfitting when only limited training samples are available [14].

To address the challenge of few-shot HSI classification, researchers have explored various solutions. Transfer learning [15], [16], [17], [18], [19] is one of the most commonly adopted methods, which improves classification performance by pretraining a model on a source domain and transferring it to the target domain. Meta-learning [20], [21] enhances performance by optimizing initial parameters through the guidance of a query set. However, these methods still rely heavily on labeled data or high-quality out-of-domain datasets. When the out-of-domain data are unavailable or mismatched with the target task, model performance often degrades significantly.

To overcome the above limitations, this article proposes PIMamba, a few-shot HSI classification framework with integrated physical constraints. By introducing physical constraints [22], the proposed framework incorporates the physical prior knowledge of HSI into the model structure, providing robust regularization and constraint mechanisms during training. Specifically, the proposed method models the evolution of hyperspectral features under physically inspired diffusion dynamics and integrates these constraints into the state-space-based Mamba architecture.

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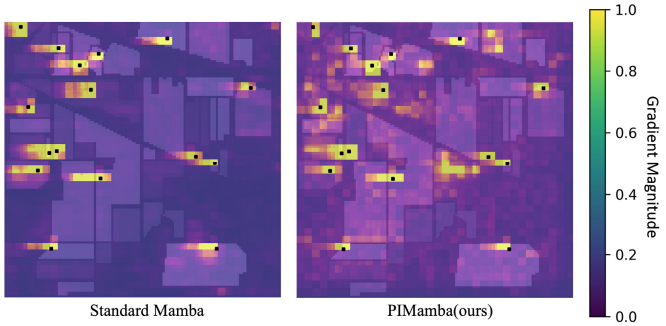


Fig. 1. Illustration of gradient contributions. Taking the IP dataset as an example, the relative gradients of the input HSI for both models are visualized at the end of training. A higher relative gradient at a point (indicated by brighter colors) signifies a greater contribution to the loss function. The black squares represent the locations of labeled training samples.

To intuitively demonstrate the effectiveness of the proposed method, we visualized the gradient contributions of each pixel to the model optimization process, as shown in Fig. 1. It can be observed that the gradients of the standard Mamba are mainly concentrated around the labeled training samples, whereas PIMamba propagates gradient information over a larger spatial area and exhibits a more uniform distribution. This indicates that the introduced physical constraints not only enhance the model's utilization of global spectral–spatial features but also enable it to effectively exploit more unlabeled information under few-shot conditions.

The main contributions of this article are as follows.

1) A physics-informed state-space modeling approach is proposed for hyperspectral feature learning. By introducing diffusion-based physical constraints into the state transition process of the Mamba architecture, the hidden-state evolution is guided by physically inspired spectral–spatial diffusion dynamics. This design enables the model to capture the intrinsic structural relationships in hyperspectral data and improves the stability of feature learning under few-shot conditions.
2) The PIMamba framework proposed in this article incorporates spectral physical loss (SpeLoss) and spatial physical loss (SpaLoss), introduces physical constraints in the feature extraction process, and significantly improves classification performance, especially in small sample scenarios. Extensive experiments demonstrate that PIMamba achieves state-of-the-art (SOTA) performance on multiple benchmark hyperspectral datasets without relying on out-of-domain annotated data or large-scale pretrained models.
3) A dynamic parameter adjustment (DPA) mechanism is designed to adaptively balance the contributions of spectral and spatial physical constraints during training. This mechanism dynamically adjusts the weights of SpeLoss and SpaLoss, allowing the model to better coordinate spectral–spatial feature learning.

II. Related Work

A. Mamba in Remote Sensing Applications

Preliminary studies have explored the application of Mamba [12] in HSI classification tasks. To address the spectral–spatial joint modeling problem in HSI classification, researchers have proposed various improved Mamba architectures. For example, Li et al. [23] developed the MambaHSI framework, which was the first to achieve image-level HSI classification based on Mamba. By incorporating a spectral-spatial fusion module, it balanced resource efficiency and classification performance. He et al. [24] proposed the 3DSS-Mamba framework, which utilized a 3-D selective scanning mechanism to capture spectral–spatial relationships, achieving SOTA performance on multiple benchmark datasets. Liu et al. [25] designed HyperMamba, which dynamically enhanced spectral features and applied neighborhood spatial scanning to achieve precise classification while effectively reducing the impact of redundant information. Additionally, Sheng et al. [26] developed a lightweight DualMamba network that combined Mamba modules with convolutional operations, enhancing spectral–spatial feature extraction capabilities while reducing computational complexity.

To address the challenge of multimodal data fusion, some studies have combined Mamba with other models to create multimodal joint classification frameworks. For example, Zeng et al. [27] proposed HLMamba, which achieved joint classification of HSI and LiDAR data through a gradient fusion algorithm and multimodal feature extraction module, demonstrating exceptional classification accuracy. Zhang et al. [28] introduced the S2CrossMamba framework, which adopts a multimodal state dynamic update mechanism, achieving significant performance improvements in multimodal HSI classification. Yang et al. [29] developed GraphMamba, which integrates Mamba with GNNs, enhancing spectral–spatial joint information modeling through linear spectral encoding and spatial context-aware modules.

To improve the adaptability and generalization ability of HSI classification in complex scenarios, some studies have focused on enhancing Mamba's dynamic properties and global modeling capabilities. Huang et al. [30] proposed dynamic token augmentation Mamba (DTAM), which improves model performance in unsupervised domain generalization scenarios through a dynamic token augmentation module. Pan et al. [31] introduced MambaLG, which combines short- and long-range spectral perception modules to achieve dynamic modeling of spectral and spatial information from local to global levels. Ahmad et al. [32] proposed WaveMamba, which combines wavelet transforms with the Mamba framework, enhancing classification performance in complex scenarios through multiresolution feature extraction.

B. Physics-Informed Neural Networks

Physics-informed neural networks (PINNs) were first proposed by Raissi et al. [22]. The core idea of PINNs is to embed physical laws, such as partial differential equations (PDEs), directly into the loss function of the neural network. This allows the neural network not only to learn from limited observational data but also to leverage physical principles to constrain the learning process.

Recent studies have explored the applications of PINNs in the field of remote sensing. For example, in wavefield simula-

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tion and seismic problems, Waheed [33] combined Kronecker neural networks with PINNs and proposed a frequency-domain wavefield modeling method to solve the Helmholtz equation. By addressing the spectral bias problem, they achieved rapid and accurate simulation of high-frequency wavefields.

In the field of electromagnetic problems, Wang et al. [34] utilized deep operator network (DeepONet) to solve 2-D electromagnetic scattering problems. By embedding physical information, they achieved efficient solutions to PDEs.

In vegetation and environmental modeling, PINNs have also demonstrated their strong expressiveness. Hossain et al. [35] combined radiative transfer models (RTMs) with PINNs to estimate nitrogen concentrations in vegetation, establishing cross-scale spectral data relationships and significantly improving prediction accuracy.

III. Proposed Methodology

A. Research Motivation

With the advancement of hyperspectral remote sensing technology, the acquisition capability of hyperspectral data has significantly improved, leading to a substantial increase in the number and scale of publicly available datasets. However, even in the era of big data, few-shot learning (FSL) remains highly relevant. On the one hand, the annotation of hyperspectral data is costly. Significant spectral response differences and domain distribution shifts often exist between different sensors, making it difficult to directly transfer existing annotated data to new application scenarios. On the other hand, in remote sensing applications, certain land cover types (e.g., rare minerals and protected vegetation) inherently have a limited number of samples, necessitating the development of FSL techniques.

Current mainstream FSL methods typically rely on large-scale pretrained models. However, pretrained models often capture only generic features independent of category, struggling to accurately characterize the fine-grained spectral differences and spatial structures within hyperspectral data. Consequently, the key research question addressed in this work is: how can we achieve effective generalization with extremely limited labeled samples without relying on large-scale pretrained models or additional out-of-domain datasets?

In recent years, PINNs have demonstrated excellent data efficiency and generalization capabilities across multiple scientific computing domains. However, unlike tasks with explicit physical equations such as wave propagation or electromagnetic scattering, HSI classification lacks directly applicable physical governing equations. This makes the construction of appropriate physical constraints a noteworthy area for exploration.

Based on this insight, this article aims to explore potential physical laws and their mathematical formulations within the unique context of HSI classification, investigating how to integrate such physical information with advanced deep learning models. By constructing a physics-guided HSI classification framework, we seek to improve classification performance and robustness under few-shot settings, offering a novel perspective on solving the few-shot HSI classification problem.

B. PIMamba Block

Mamba is a recently proposed deep learning framework based on the SSM, designed for efficient modeling of high-dimensional sequential data. SSMs describe the dynamic relationship between inputs and outputs through a hidden-state vector. The continuous formulation can be written as the following differential equations:

$$\begin{aligned} \frac{d}{dt}h(t) &= Ah(t) + Bx(t) \\ y(t) &= Ch(t) \end{aligned} \tag{1}$$

where $A \in \mathbb{R}^{N\times N}$ denotes the state transition matrix, $h(t) \in \mathbb{R}^N$ represents the hidden state of the system, $B \in \mathbb{R}^{N\times M}$ denotes the input projection matrix, $x(t) \in \mathbb{R}^M$ is the input signal of the system, $C \in \mathbb{R}^{P\times N}$ represents the output projection matrix, and $y(t) \in \mathbb{R}^P$ is the output signal.

The state transition matrix A determines the propagation speed, decay behavior, and stability of the system. Therefore, the structure of A largely determines the dynamic patterns that the model can learn. In the original Mamba model, the matrix A is typically parameterized as a fully learnable matrix, whose dynamical structure is entirely learned from data. However, in high-dimensional spatial data such as remote sensing imagery, hidden states usually exhibit strong local correlations. Adjacent pixels often share similar spectral and semantic properties. Such spatial continuity implies that the evolution of hidden states tends to exhibit local propagation and diffusion behaviors.

Therefore, in PIMamba, we model the hidden-state evolution as a continuous-time system constrained by physical diffusion

$$\frac{d}{dt}h(t) = -L_\phi h(t) + Bx(t) \tag{2}$$

where $L_\phi \in \mathbb{R}^{N\times N}$ denotes the physical diffusion operator, which describes the local propagation and attenuation of hidden states in the feature space. The negative sign is introduced because the classical form of diffusion systems is $h'(t) = -Lh(t)$, where L is typically a positive semidefinite operator. This formulation guarantees that the total energy of the system decays or converges toward equilibrium over time, thereby naturally ensuring system stability. The physical diffusion operator L_ϕ is defined as

$$L_\phi = \alpha D + \beta I \tag{3}$$

where $D \in \mathbb{R}^{N\times N}$ denotes the discrete diffusion operator, I is the identity matrix, and α and β are the nonnegative physical parameters that control the diffusion strength and attenuation strength, respectively. The operator D is used to characterize the local interaction relationships among hidden states, and its specific form can be defined according to the modeling target of different branches.

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For the spectral branch, the hidden states are arranged along the spectral dimension. Therefore, a 1-D second-order difference operator is adopted as D, given by

$$D = \begin{bmatrix} -2 & 1 & 0 & \cdots & 0 \\ 1 & -2 & 1 & \cdots & 0 \\ 0 & 1 & -2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & 1 \\ 0 & 0 & 0 & 1 & -2 \end{bmatrix}. \tag{4}$$

For the spatial branch, the hidden states correspond to spatial sequences on the image plane. Therefore, a 2-D discrete Laplacian operator can be adopted as D to characterize the local coupling and propagation relationships between neighboring spatial locations.

To adapt the model to discrete-time scenarios and improve computational efficiency, the continuous system is discretized using the zero-order hold (ZOH) method. Under the discrete time step Δ_k corresponding to the kth input position, the state transition matrix and the input matrix can be expressed as

$$\begin{aligned} \bar{A}_k &= \exp\left(-\Delta_k L_\phi\right) \\ \bar{B}_k &= \left(-\Delta_k L_\phi\right)^{-1}\left(I - \bar{A}_k\right) \cdot \Delta_k B_k. \end{aligned} \tag{5}$$

Thus, the discrete state update equations can be written as

$$\begin{aligned} h_k &= \bar{A}_k h_{k-1} + \bar{B}_k x_k \\ y_k &= C h_k. \end{aligned} \tag{6}$$

Furthermore, to preserve the input-adaptive property of PIMamba, the diffusion coefficients α and β, and the time step Δ_k are dynamically generated from the input features

$$\begin{aligned} \alpha &= \tau\left(\mathrm{Proj}_\alpha\left(x_k\right)\right) \\ \beta &= \tau\left(\mathrm{Proj}_\beta\left(x_k\right)\right) \\ \Delta_k &= \tau\left(\mathrm{Proj}_\Delta\left(x_k\right)\right) \end{aligned} \tag{7}$$

where $\mathrm{Proj}_\alpha(\cdot)$, $\mathrm{Proj}_\beta(\cdot)$, and $\mathrm{Proj}_\Delta(\cdot)$ denote the linear projection functions, and $\tau(\cdot)$ represents the Softplus activation function, which ensures that the physical parameters remain positive.

Through the above design, the state transition matrix in PIMamba is no longer a completely freely learned parameter, but is determined by the physical diffusion operator L_ϕ. This allows the hidden-state evolution to satisfy stable physical constraints, thereby enabling more robust feature representations under small-sample and complex noise conditions.

C. Proposed PIMamba Framework

In the task of HSI classification, spectral–spatial joint analysis [36] has been widely demonstrated as a crucial approach for improving accuracy and robustness. As shown in Fig. 2, the proposed PIMamba framework incorporates two parallel branches: the spatial branch and the spectral branch. Both branches utilize the Mamba block to extract serialized features, differing only in how their respective sequences are constructed.

To better capture the spatial distribution characteristics of pixels, the spatial branch treats each pixel as a sequence. For an HSI with a resolution of $H \times W \times C$, all pixels are treated as a sequence of length $T_{\mathrm{spa}} = H \times W$. In the Mamba block, each timestep in the sequence corresponds to the channel vector of a pixel, with a dimension of C. The network performs state iterations on this sequence, layer by layer, aggregating local and global spatial information. In the next stage, average pooling is applied to reduce the sequence length, thereby fusing higher level features.

To enable the network to model the "spectral dimension" directly, the spectral branch treats the channels themselves as a sequence, allowing the Mamba block to extract features that capture potential patterns among spectral bands. To avoid excessive input dimensions caused by the large number of channels, the spectral branch divides the C channels into N groups, each containing C/N channels. Thus, each time step corresponds to a group of channels. The input tensor (B, H, W, C) is transformed through preprocessing into a batch of size $B \times H \times W$ with sequence length N and feature dimension C/N. By iteratively updating the hidden states along the spectral sequence, the spectral branch learns finer spectral features and differences between spectral bands.

Another core aspect of the PIMamba model is the introduction of SpaLoss and SpeLoss based on physical priors. Since the output feature map of the spatial branch primarily reflects the dependencies between pixels in the 2-D plane of the image, it is used to construct the spatial physical constraint. In contrast, the feature map extracted by the spectral branch focuses on the intrinsic relationships between spectral bands, serving as the basis for measuring the spectral physical constraint.

D. Spectral Physical Loss

In hyperspectral data analysis, we focus on the distribution of spectra along the wavelength axis λ. A large body of literature and experiments [37], [38], [39], [40], [41] have observed that spectral distributions generally exhibit the following two physical phenomena.

1) *Reflectance tends to be smoother between neighboring bands of the spectrum.*

In actual spectral measurements, when a narrowband light source with a wavelength of λ_0 enters the instrument, the measured signal will be distributed over a small range of neighboring wavelengths rather than being confined to λ_0 alone. From the instrument's perspective, the output curve of the response to monochromatic light is referred to as the instrument response function (IRF). The relationship can be described as

$$I_{\mathrm{meas}}(\lambda) = \int_{-\infty}^{\infty} I_{\mathrm{true}}(\lambda_0) \cdot IRF(\lambda - \lambda_0)\, d\lambda_0 \tag{8}$$

where $IRF(\lambda - \lambda_0)$ is centered at λ_0, representing the response weight at λ. Therefore, the true photon signal I_{true} incident on the detector is convolved with the instrument response function, so that the measured spectrum I_{meas} does not exhibit abrupt jumps between neighboring bands but instead shows a certain degree of smooth variation. This "convolutional smoothing" is very similar in nature to the diffusion process: it locally smoothes out excessively sharp differences and

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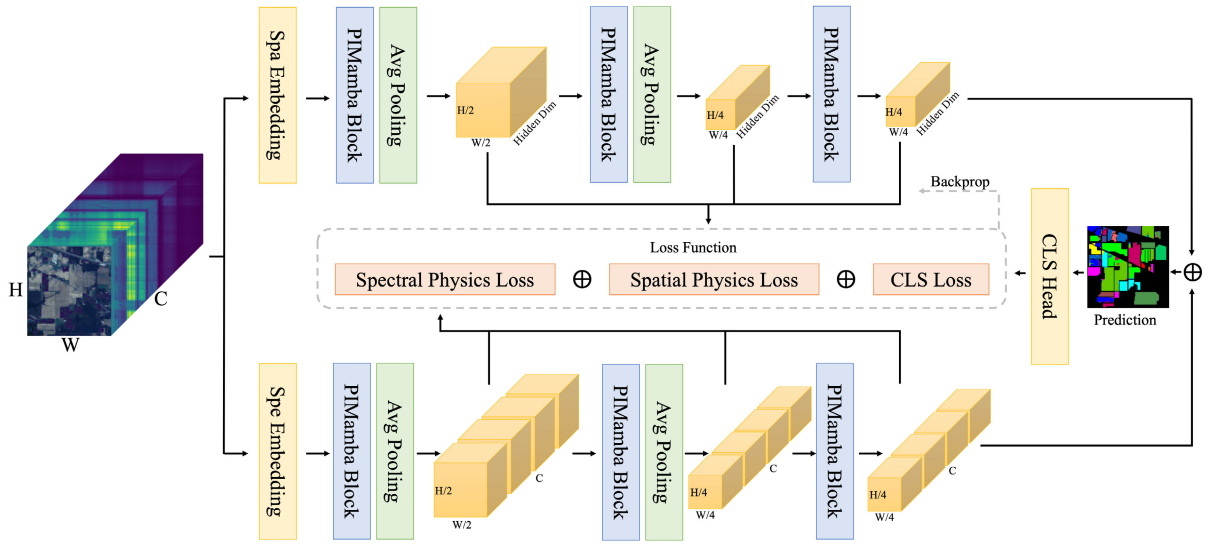


Fig. 2. Framework of the proposed PIMamba. The framework consists of a spectral branch and a spatial branch, both built upon the proposed PIMamba blocks for sequence modeling. The spectral branch maintains the number of spectral channels C in the feature maps across all stages. The feature maps in the spectral branch contribute to the SpeLoss, while the feature maps in the spatial branch contribute to the SpaLoss. After the feature extraction stage, the two branches are fused and fed into the classification head to produce the prediction map, which provides the classification loss.

encourages the function to be continuous or gently varying between neighboring bands.

2) *Absorption or enhancement peaks and valleys occur at specific wavelengths of the spectrum.*

In the process of hyperspectral imaging, the absorption intensity of light at different wavelengths varies depending on the material type. The reduction in intensity caused by absorption in a homogeneous medium can be described by the Beer–Lambert law, a fundamental principle applicable to all electromagnetic radiation and all materials. The law is mathematically expressed as

$$I(\lambda) = I_0(\lambda) \cdot e^{-k(\lambda)d} \tag{9}$$

where $I(\lambda)$ represents the transmitted intensity, $I_0(\lambda)$ represents the incident intensity, $k(\lambda)$ represents the absorption coefficient, and d represents the optical path length. When $k(\lambda)$ is significantly high at a certain λ_0, the energy at this wavelength decreases sharply, leading to absorption valleys in the measured spectra.

The absorption coefficient $k(\lambda)$ often increases sharply at certain wavelengths, correlating with the resonant frequencies or energy-level differences of molecules or particles. For example, photon frequency $v = c/\lambda$ determines whether it matches the vibrational energy levels of particles or molecules. Common examples include strong red/blue light absorption (chlorophyll effects) in vegetation spectra, high near-infrared reflectance, and strong infrared absorption in water spectra.

Based on the above two physical phenomena, the physical constraints in characterizing the spectral curve should be both overall smoothness and allow for absorption at certain wavelengths. By treating the spectral reflectance as a result of both diffusion (smoothness) and absorption (reaction) mechanisms, the evolution of the spectrum can be characterized by an approximate "reaction–diffusion equation." Based on this, we abstract the following 1-D PDE to approximate the change patterns of features in the spectral dimension:

$$\frac{\partial R}{\partial \lambda}(\lambda) - \alpha \frac{\partial^2 R}{\partial \lambda^2}(\lambda) + \beta R(\lambda) = 0 \tag{10}$$

where $R(\lambda)$ represents the variation function of spectral reflectance with respect to the wavelength λ; α is the "diffusion coefficient," reflecting the smoothing intensity of spectral bands; and β is the "reaction coefficient," used to capture localized absorption, reflection, and other physical component-related variations within specific bands. The existence of $\partial^2 R/\partial \lambda^2$ enforces constraints on the curvature of the spectral curve, encouraging its continuity and stability across adjacent bands.

In practical scenarios, the wavelength λ is discretized into bands, with each band corresponding to a central wavelength λ_i. Therefore, in actual calculations, the terms $\partial R/\partial \lambda$ and $\partial^2 R/\partial \lambda^2$ are approximated as

$$\begin{aligned} \frac{\partial R}{\partial \lambda}(\lambda_i) &\approx \frac{R(\lambda_{i+1}) - R(\lambda_i)}{\Delta\lambda} \\ \frac{\partial^2 R}{\partial \lambda^2}(\lambda_i) &\approx \frac{R(\lambda_{i+1}) - 2R(\lambda_i) + R(\lambda_{i-1})}{(\Delta\lambda)^2} \end{aligned} \tag{11}$$

where $\Delta\lambda$ denotes the interval between adjacent bands. Based on (13) and (14), the PDE residual at band i can be computed as

$$f_{\text{Spe}}(\lambda_i) = \frac{R_{i+1} - 2R_i + R_{i-1}}{(\Delta\lambda)^2} - \alpha \frac{R_{i+1} - R_i}{\Delta\lambda} + \beta R_i. \tag{12}$$

For simplicity, let $R(\lambda_i)$ be denoted as R_i.

For the spectral branch, we define the SpeLoss $\mathcal{L}_{\text{Spe}}$ as

$$\mathcal{L}_{\text{Spe}} = \sum_{\text{pixel}} \sum_{i=1}^{C} \left(f_{\text{Spe}}(\lambda_i)\right)^2 \tag{13}$$

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where C represents the number of discretized spectral bands, and pixel denotes all image pixels. During training, minimizing $\mathcal{L}_{\text{Spe}}$ encourages the network to satisfy the physical priors derived from the reaction–diffusion framework along the spectral dimension.

E. Spatial Physical Loss

Spatial structures in HSI also contain significant physical information. Numerous studies [42], [43] have demonstrated two prominent physical phenomena in spatial structures.

1) *Spatially neighboring pixels of the same class tend to have similarities.*

Due to the continuity of material coverage, pixels within the same region—such as forests, farmland, or urban areas—tend to show minimal variations between neighboring pixels. This phenomenon can be observed across various types of remote sensing images, including HSI, synthetic aperture radar (SAR), and RGB.

2) *Localized areas tend to be smoother, with relatively abrupt edges or feature areas.*

Pixel values (e.g., reflectance or feature vectors) are continuous or quasi-continuous across neighboring coordinates (x, y), with significant variations occurring only at true boundaries or feature zones, such as transitions between material types, or at the edges of roads or water bodies.

Based on these two physical phenomena, the interaction among adjacent pixels in the spatial dimension, which leads to local smoothness, can be regarded as an analogy to the diffusion phenomenon: just as heat or concentration balances locally and tends to smooth out, pixel values in an image also "converge" or "reduce noise" within a small region. Leveraging these physical characteristics, we designed spatial branch-specific physical constraints.

In the spatial branch, the 2-D spatial dimensions (row and column coordinates of the image) are used to describe the interactions between pixels in hyperspectral data. The feature distribution of a specific point in an HSI is denoted as $F(x, y)$, where x and y represent the horizontal and vertical coordinates of the image, respectively. To model the above physical properties, we use the following 2-D diffusion PDE:

$$\frac{\partial F}{\partial t} = D_x \frac{\partial^2 F}{\partial x^2} + D_y \frac{\partial^2 F}{\partial y^2} - R(x, y) \tag{14}$$

where D_x and D_y represent the diffusion coefficients in the horizontal and vertical directions, respectively, reflecting the interaction intensity between adjacent pixels. $R(x, y)$ represents the source term used to describe the temporal evolution of the local characteristics of specific objects (e.g., boundary features or hotspots). During training, we assume a steady state where $\partial F/\partial t = 0$.

When performing calculations on a discrete grid (pixel grid), we use the finite difference method for discretization

$$\begin{aligned} \frac{\partial^2 F}{\partial x^2}(i, j) &\approx \frac{F_{i+1,j} - 2F_{i,j} + F_{i-1,j}}{(\Delta s)^2} \\ \frac{\partial^2 F}{\partial y^2}(i, j) &\approx \frac{F_{i,j+1} - 2F_{i,j} + F_{i,j-1}}{(\Delta s)^2} \end{aligned} \tag{15}$$

where $F_{i,j}$ represents the feature value at pixel (i, j), and Δs represents the grid spacing in the horizontal and vertical directions. Based on (17) and (18), the PDE residual at pixel (i, j) can be computed as

$$\begin{aligned} f_{\text{spa}}(i, j) = D_x &\frac{F_{i+1,j} - 2F_{i,j} + F_{i-1,j}}{(\Delta s)^2} \\ &+ D_y \frac{F_{i,j+1} - 2F_{i,j} + F_{i,j-1}}{(\Delta s)^2} - R_{i,j}. \end{aligned} \tag{16}$$

For the spatial branch, we define the SpaLoss $\mathcal{L}_{\text{spa}}$ as

$$\mathcal{L}_{\text{spa}} = \sum_{\lambda=1}^{C} \left(\frac{1}{N_x N_y} \sum_{i=1}^{N_x} \sum_{j=1}^{N_y} \left(f_{\text{spa}}(i, j) \right)^2 \right). \tag{17}$$

The objective of this loss term is to minimize the residual $f_{\text{spa}}(i, j)$, ensuring that the network learns features consistent with the physical law of spatial smoothness and local uniformity.

F. Parameter Determination Method

For the data-driven classification loss $\mathcal{L}_{\text{cls}}$, we adopt the commonly used cross-entropy loss function. Based on this, the total loss can be written as

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + w_1 \mathcal{L}_{\text{spe}} + w_2 \mathcal{L}_{\text{spa}}. \tag{18}$$

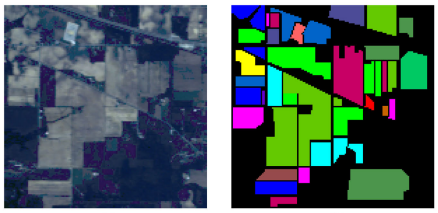
The determination of the unknown parameters w_1 and w_2 is critically important. They control the weights of the SpeLoss $\mathcal{L}_{\text{spe}}$ and the SpaLoss $\mathcal{L}_{\text{spa}}$ in the total loss, determining the balance between the supervised signal and physical constraints. Besides w_1 and w_2, other parameters in $\mathcal{L}_{\text{spe}}$, such as the diffusion coefficient α and reaction coefficient β, and in $\mathcal{L}_{\text{spa}}$, such as the horizontal diffusion coefficient D_x and vertical diffusion coefficient D_y, are also crucial. To reasonably determine these weights, we adopt a DPA mechanism. In the following, we illustrate this mechanism using w_1 and w_2 as examples.

Algorithm 1 DPA Mechanism

Input: Network parameters θ, Loss weights w_1, w_2
for $e = 1$ **to** *Epochs* **do**
  Forward pass for the current epoch
  Compute $\mathcal{L}_{\text{spe}}$ and $\mathcal{L}_{\text{spa}}$
  Compute gradient norms
  $g_{\text{spe}} \leftarrow \left\| \frac{\partial \mathcal{L}_{\text{spe}}}{\partial \theta} \right\|_2$
  $g_{\text{spa}} \leftarrow \left\| \frac{\partial \mathcal{L}_{\text{spa}}}{\partial \theta} \right\|_2$
  Gradient normalization to balance each loss
  $w_1 \leftarrow w_1 \times \sqrt{\frac{g_{spa}+\epsilon}{g_{spe}+\epsilon}}$
  $w_2 \leftarrow w_2 \times \sqrt{\frac{g_{spe}+\epsilon}{g_{spa}+\epsilon}}$
  Compute total loss with updated weights
  $\mathcal{L}_{\text{total}} = w_1 \times \mathcal{L}_{\text{spe}} + w_2 \times \mathcal{L}_{\text{spa}}$
  Backpropagation and parameter update
  $\theta \leftarrow \theta - \eta \times \frac{\partial L_{\text{total}}}{\partial \theta}$
end for
Output: Optimized network parameters θ, Updated weights w_1, w_2

The DPA mechanism optimizes these parameters through iteration. For the total loss $\mathcal{L}_{\text{total}}$ in (18), we apply the gradient normalization (GN) method. The core idea of GN

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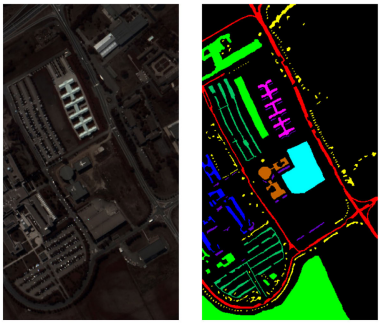


(a) (b)

ID	Class	Color	Samples
1	Alfalfa		46
2	Corn-notill		1428
3	Corn-mintill		830
4	Corn		237
5	Grass-pasture		483
6	Grass-trees		730
7	Grass-pasture-mowed		28
8	Hay-windrowed		478
9	Oats		20
10	Soybean-notill		972
11	Soybean-mintill		2455
12	Soybean-clean		593
13	Wheat		205
14	Woods		1265
15	Buildings-Grass-Trees-Drives		386
16	Stone-Steel-Towers		93

(c)

Fig. 3. IP dataset. (a) False color image. (b) Ground truth. (c) Class name with samples.



(a) (b)

ID	Class	Color	Samples
1	Asphalt		6631
2	Meadows		18649
3	Gravel		2099
4	Trees		3064
5	Painted-metal-sheets		1345
6	Bare-Soil		5029
7	Bitumen		1330
8	Self-Blocking-Bricks		3682
9	Shadows		947

(c)

Fig. 4. UP dataset. (a) False color image. (b) Ground truth. (c) Class name with samples.

is to adjust the weight of each loss term by balancing the gradient paradigm of each loss. The detailed steps are shown in Algorithm 1. First, we calculate the gradient norms of the SpeLoss $\mathcal{L}_{\text{spe}}$ and SpaLoss $\mathcal{L}_{\text{spa}}$ with respect to the network parameters θ

$$g_{\text{spe}} = \left\| \frac{\partial \mathcal{L}_{\text{spe}}}{\partial \theta} \right\|_2, \quad g_{\text{spa}} = \left\| \frac{\partial \mathcal{L}_{\text{spa}}}{\partial \theta} \right\|_2. \tag{19}$$

To balance the contributions of two branches to the total loss, the parameters w_1 and w_2 need to satisfy the relationship $w_1 g_{spa} = w_2 g_{spe}$. If w_1 and w_2 are updated directly according to this linear relationship, when g_{spe} and g_{spa} differ greatly, it will lead to drastic fluctuations in the parameters. Therefore, we add a square root to the parameter update formula

$$w_1 \leftarrow w_1 \cdot \sqrt{\frac{g_{spa} + \epsilon}{g_{spe} + \epsilon}}, \quad w_2 \leftarrow w_2 \cdot \sqrt{\frac{g_{spe} + \epsilon}{g_{spa} + \epsilon}} \tag{20}$$

where ϵ is a small positive number used to avoid a zero denominator. According to the update rule of (20), w_1 and w_2 are dynamically adjusted in each backpropagation to always maintain an appropriate proportion of the spectral and spatial losses in the total loss.

IV. Experiment

A. Datasets

In this article, we selected four widely used datasets in HSI classification: Indian Pines (IP), University of Pavia (UP), Salinas Scene (SA), and Houston 2013 (Houston).

IP: This dataset covers the agricultural region of northern Indiana, USA. It includes 16 land cover classes, with the sample distribution shown in Fig. 3. The spatial resolution of the dataset is 20 m, with dimensions of 145×145 pixels. The raw data contain 224 spectral bands. To minimize the impact of water absorption bands on classification performance, 200 effective spectral bands are typically retained.

UP: This dataset covers the UP in Italy. This dataset is a typical urban hyperspectral dataset. It includes nine land cover classes, with sample numbers shown in Fig. 4. The dataset has a high spatial resolution of 1.3 m and an image size of 610×340 pixels. After preprocessing to remove noise-affected bands, 103 effective channels are retained.

SA: This dataset covers California's Salinas Valley. It is predominantly agricultural and contains 16 land cover classes, whose sample details appear in Fig. 5. The spatial resolution is 3.7 m, with an overall size of 512×217 pixels. After preprocessing, 204 channels were retained.

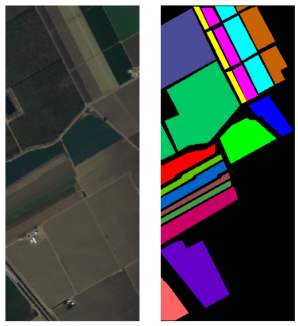
Houston: This dataset covers the University of Houston along with its surrounding urban region in the USA. It comprises 15 land cover categories, with sample counts provided in Fig. 6. The imagery has a spatial resolution of 2.5 m, covering an area of 349×1905 pixels, and includes a total of 144 spectral bands.

B. Experimental Details

All experiments were carried out within a unified setup running Ubuntu 22.04, configured with PyTorch 2.5.1, Python 3.12, and compatible with CUDA 12.4. The hardware consisted of an RTX 4090 (24 GB) GPU along with a 16-core Intel[1] Xeon[1] Gold 6430 CPU. In order to strengthen result consistency and reliability, we performed each experiment ten times, recording both the average outcome and standard deviation. In FSL experiments, we employed a rigorous evaluation protocol where only 5 pixel samples per class were randomly

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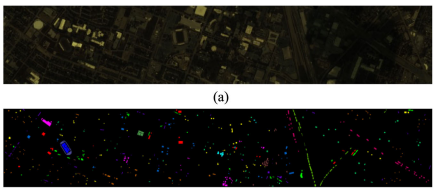


(a) (b)

ID	Class	Color	Samples
1	Brocoli-green-weeds-1		2009
2	Brocoli-green-weeds-2		3726
3	Fallow		1976
4	Fallow-rough-plow		1394
5	Fallow-smooth		2678
6	Stubble		3959
7	Celery		3579
8	Grapes-untrained		11271
9	Soil-vineyard-develop		6203
10	Corn-senesced-green-weeds		3278
11	Lettuce-romaine-4wk		1068
12	Lettuce-romaine-5wk		1927
13	Lettuce-romaine-6wk		916
14	Lettuce-romaine-7wk		1070
15	Vinyard-untrained		7268
16	Vinyard-vertical-trellis		1807

(c)

Fig. 5. SA dataset. (a) False color image. (b) Ground truth. (c) Class name with samples.



(a)

(b)

ID	Class	Color	Samples
1	Healthy-Grass		1251
2	Stressed-Grass		1254
3	Synthetic-Grass		697
4	Tree		1244
5	Soil		1242
6	Water		325
7	Residential		1268
8	Commercial		1244
9	Road		1252
10	Highway		1227
11	Railway		1235
12	Parking-Lot-1		1233
13	Parking-Lot-2		469
14	Tennis-Court		428
15	Running-Track		660

(c)

Fig. 6. Houston dataset. (a) False color image. (b) Ground truth. (c) Class name with samples.

selected for training from each of the four datasets, with all remaining samples reserved for testing.

C. Evaluation Metrics

To comprehensively evaluate the classification performance, we adopt overall accuracy (OA), average accuracy (AA), the Kappa coefficient (Kappa), the macro-averaged $F1$-score (mF1), and computational efficiency indicators, including the number of parameters (Params) and floating-point operations (FLOPs).

D. Comparison Methods

We selected eight high-performance methods that do not rely on pretrained models or external datasets. These include CNN-based methods: 3DCNN and HybridSN; Transformer-based methods: SpectralFormer, spectral–spatial feature tokenization transformer (SSFTT), and groupwise separable convolutional vision transformer (GSC-ViT); and Mamba-based methods: MambaHSI, S^{2}Mamba, and GraphMamba.

1) *3DCNN [4] (TGRS 2018):* The 3DCNN simultaneously processes spectral and spatial information, enabling higher classification accuracy and lower computational costs.
2) *HybridSN [5] (GRSL 2020):* HybridSN combines 3D-CNN and 2D-CNN in a hybrid network model, effectively balancing classification accuracy and computational complexity.
3) *SpectralFormer [8] (TGRS 2022):* The SpectralFormer model focuses on extracting local sequential information from adjacent bands in hyperspectral data, significantly improving classification performance.
4) *SSFTT [9] (TGRS 2022):* SSFTT employs feature tokenizers and Transformer encoder modules to efficiently capture low-level and high-level spectral and spatial features.
5) *GSC-ViT [10] (TGRS 2024):* GSC-ViT introduces a group-separated multihead self-attention mechanism (GSSA) to effectively capture spectral and spatial features.
6) *MambaHSI [23] (TGRS 2024):* MambaHSI is the first hyperspectral classification method based on the Mamba model. It achieves efficient global modeling and precision by integrating spectral and spatial feature blocks.
7) *S^{2}Mamba [44] (TGRS 2025):* S^{2}Mamba features a bidirectional scanning mechanism and dynamic signature competition capabilities.
8) *GraphMamba [29] (TGRS 2024):* GraphMamba integrates adaptive graph convolutions with selective sequence modeling, achieving breakthroughs in both accuracy and computational efficiency.

E. Experimental Results

The quantitative results on the IP dataset are presented in Table I. PIMamba achieves the best overall performance across OA, AA, Kappa, and mF1, with an OA of 79.79% (a 5.66% improvement over MambaHSI). For challenging categories such as “Soybean-notill” and “Woods,” PIMamba achieves 77.02% and 92.45% accuracy, respectively. As shown in Fig. 7, it also produces classification maps with sharper boundaries and more consistent region divisions, validating its effectiveness in few-shot HSI classification.

For the UP dataset, the quantitative classification results are presented in Table II. PIMamba achieves the best performance across all four metrics, with an OA of 85.37%,

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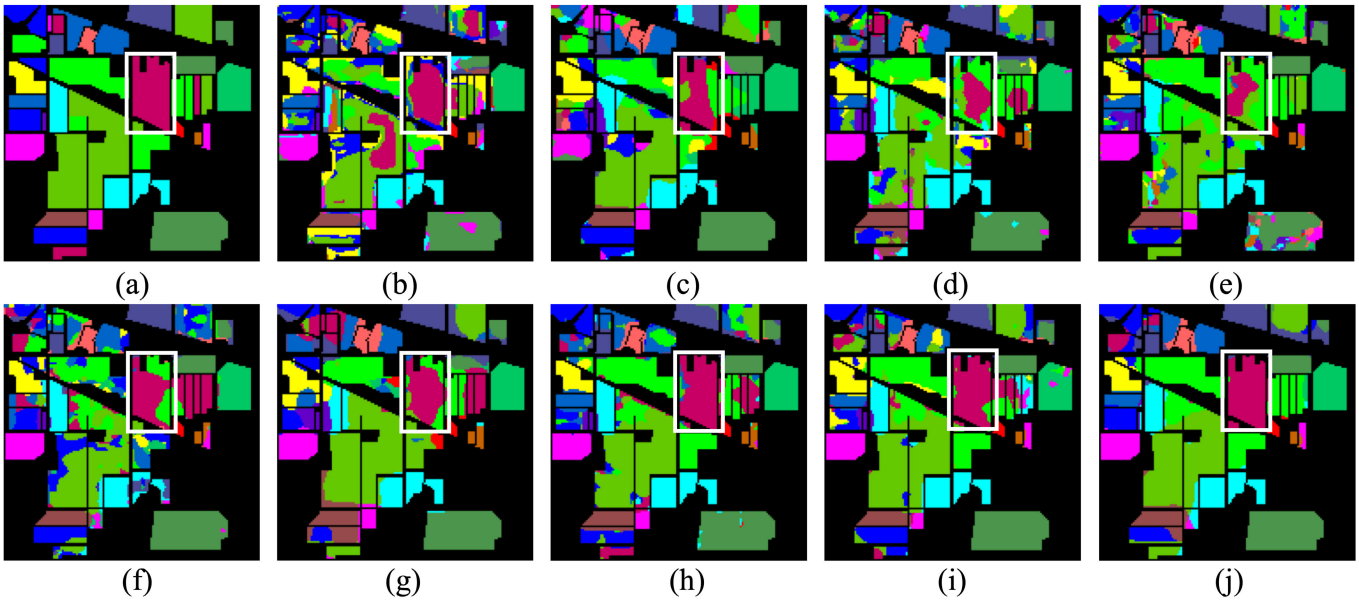


Fig. 7. Qualitative visualization of the classification map for the IP dataset. (a) Ground truth. (b) 3DCNN. (c) HybridSN. (d) SpectralFormer. (e) SSFTT. (f) GSC-ViT. (g) MambaHSI. (h) S^2Mamba. (i) GraphMamba. (j) PIMamba (ours).

TABLE I

QUANTITATIVE RESULT (ACC% ± STD%) OF IP DATASET (FIVE LABELED SAMPLES PER CLASS FOR TRAINING). THE HIGHEST VALUES ARE MARKED IN BOLD, AND SUBSEQUENT TABLES WILL FOLLOW THE SAME PATTERN

Class	3DCNN	HybirdSN	SpectralFormer	SSFTT	GSC-ViT	MambaHSI	S^2Mamba	GraphMamba	PIMamba
Alfalfa	93.87±4.88	99.35±1.29	97.42±1.94	97.1±2.26	97.42±1.94	**99.03±1.48**	96.67±4.22	96.67±3.94	98.93±2.29
Corn-notill	39.06±12.46	52.82±10.68	57.03±11.94	56.47±11.81	54.18±14.56	58.85±9.84	56.24±15.35	58.25±11.88	**72.82±5.99**
Corn-mintill	40.53±10.53	50.25±10.74	52.5±9.56	52.4±9.83	52.86±8.66	51.09±12.67	56.3±7.34	58.27±7.85	**59.63±13.13**
Corn	77.52±15.01	87.57±8.59	91.58±6.91	91.67±7.02	92.43±5.92	90.95±4.6	92.44±8.1	93.21±7.15	**93.38±6.44**
Grass-pasture	59.68±14.84	72.86±10.37	82.35±4.88	82.41±3.0	80.32±6.42	81.32±5.2	84.09±4.69	82.36±5.37	**86.26±4.44**
Grass-trees	71.1±8.62	73.23±10.0	92.71±4.0	91.43±5.15	91.75±4.76	86.63±5.96	93.81±3.53	**96.36±1.88**	91.25±4.61
Grass-pasture-mowed	100.0±0.0	100±0.0	100.0±0.0	100±0.0	100.0±0.0	100.0±0.0	99.17±2.5	99.17±2.5	**100±0.0**
Hay-windrowed	91.71±4.04	97.9±3.13	98.47±2.0	98.44±1.93	97.49±2.76	99.29±1.35	97.88±2.39	95.39±5.37	**99.43±0.82**
Oats	100.0±0.0	98.0±6.0	100.0±0.0	100±0.0	98.0±6.0	100.0±0.0	100.0±0.0	95.0±10.0	**100.0±0.0**
Soybean-notill	55.52±6.85	63.12±5.52	60.19±15.75	61.24±15.71	59.46±12.92	68.8±11.48	72.51±12.83	71.29±12.69	**77.02±5.91**
Soybean-mintill	50.61±12.12	64.68±7.82	58.91±13.8	61.14±12.59	62.85±9.03	70.23±6.29	68.82±9.51	71.19±7.57	**75.15±7.28**
Soybean-clean	48.03±10.52	49.98±10.76	57.79±12.77	57.58±12.37	60.69±9.54	62.7±9.61	**69.36±10.52**	67.26±12.04	66.43±11.28
Wheat	90.79±9.56	95.74±3.5	99.79±0.48	99.68±0.54	99.74±0.63	94.16±5.36	98.99±1.56	99.68±0.79	**99.75±0.96**
Woods	81.41±7.34	86.22±5.62	87.4±8.21	88.43±7.99	89.67±6.67	91.11±3.99	91.77±3.32	90.94±5.44	**92.45±4.51**
Buildings-Grass-Trees	77.68±8.44	82.53±12.44	94.04±6.46	94.04±6.56	93.88±6.17	90.3±8.72	94.62±7.69	**94.65±4.35**	96.17±2.89
Stone-Steel-Towers	96.54±3.93	91.54±10.32	99.36±0.64	**99.62±0.59**	99.23±1.54	95.51±4.41	99.61±0.6	99.35±0.87	99.07±1.98
OA(%)	59.06±3.44	68.19±3.12	70.39±5.23	70.98±4.57	71.16±3.79	74.13±2.18	75.54±3.54	76.23±3.23	**79.79±2.59**
AA(%)	73.38±2.13	79.11±2.0	83.1±2.37	83.23±2.11	83.12±2.12	83.75±1.05	85.77±1.33	85.56±2.13	**87.62±1.8**
Kappa(%)	71.6±2.97	77.0±3.89	80.73±2.87	80.47±2.83	80.87±2.23	79.73±3.69	82.13±3.82	83.93±3.41	**87.0±2.89**
mF1(%)	58.68±3.69	65.46±2.35	67.78±3.97	68.78±3.58	68.44±2.97	71.3±2.99	72.32±3.45	72.3±3.81	**73.46±3.16**

representing a 4.15% improvement over the standard MambaHSI model. Significant improvements were observed in the "Grave" and "Self-Blocking-Bricks" categories. The visualization of classification maps is shown in Fig. 8, where PIMamba demonstrates better continuity and sharper boundaries in highlighted areas (Self-Blocking-Bricks).

For the SA dataset, Table III shows that PIMamba achieves the best overall performance across all four metrics, with an OA of 95.52%, a 3.21% improvement over MambaHSI. In challenging classes such as "Grapes-untrained" and "Vinyard-untrained," PIMamba attains 87.47% and 92.66% accuracy, outperforming other methods. As shown in Fig. 9, it also preserves finer boundary continuity and spatial details in complex farmland and vegetation regions.

For Houston dataset, Table IV shows that PIMamba achieves the best overall performance across all four metrics, with an OA of 80.8%, a 4.06% improvement over the baseline. Notable gains are observed in challenging classes

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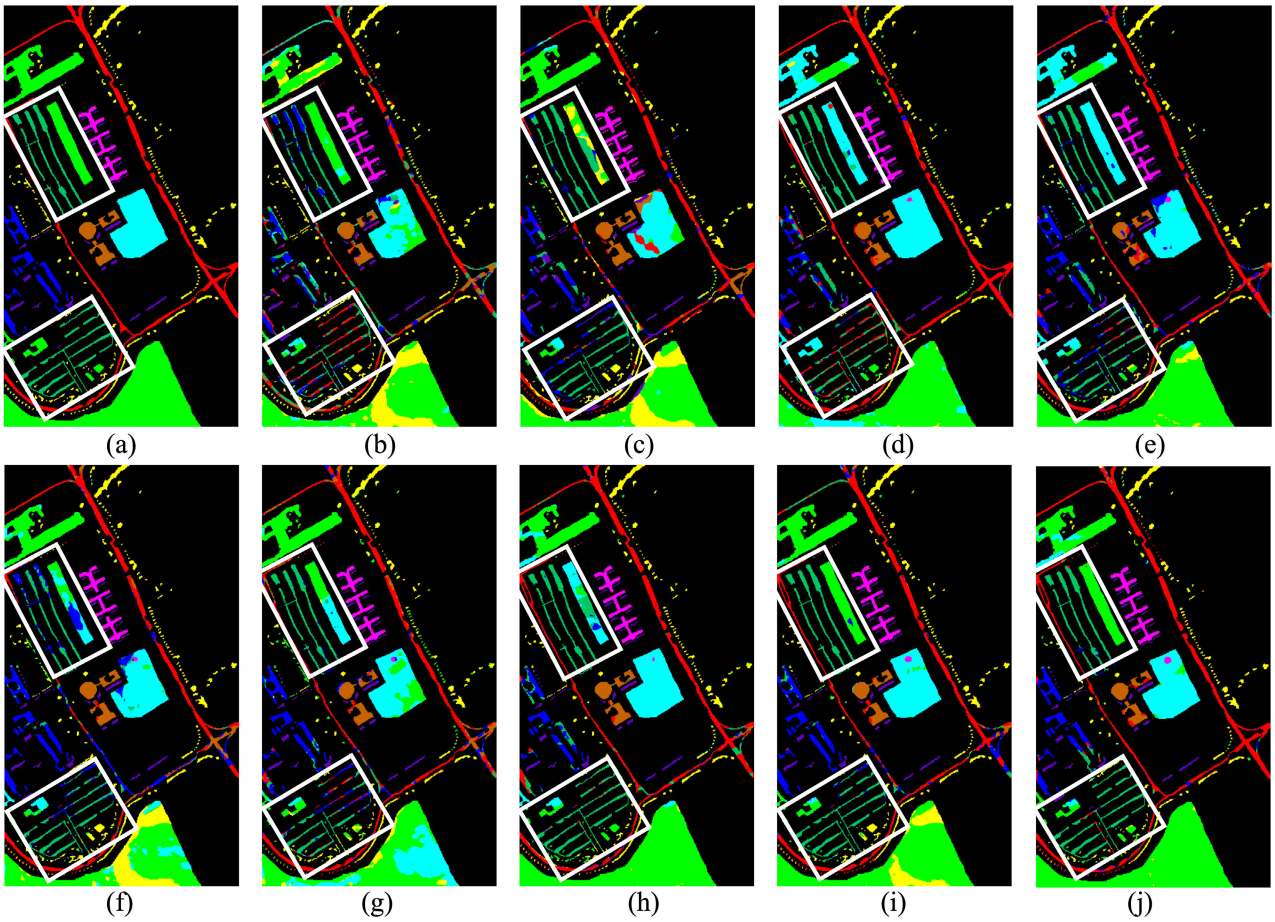


Fig. 8. Qualitative visualization of the classification map for the UP dataset. (a) Ground truth. (b) 3DCNN. (c) HybridSN. (d) SpectralFormer. (e) SSFTT. (f) GSC-ViT. (g) MambaHSI. (h) S^{2}Mamba. (i) GraphMamba. (j) PIMamba (ours).

TABLE II
QUANTITATIVE RESULT (ACC% ± STD%) OF THE UP DATASET (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	3DCNN	HybirdSN	SpectralFormer	SSFTT	GSC-ViT	MambaHSI	S^{2}Mamba	GraphMamba	PIMamba
Asphalt	60.49±9.34	63.23±14.84	78.54±13.25	78.09±13.88	80.54±10.3	78.74±12.01	73.65±10.57	81.6±13.67	**87.21±4.62**
Meadows	76.19±11.49	75.61±6.05	73.2±7.25	75.11±7.83	76.43±11.26	80.22±9.01	**83.66±7.32**	78.18±16.05	80.98±10.59
Grave	50.56±9.57	59.37±13.05	68.71±12.88	69.97±14.26	66.35±17.25	76.76±13.75	83.75±8.54	79.38±14.15	**86.28±7.15**
Trees	**87.74±8.08**	62.05±16.49	76.34±14.06	75.08±14.26	77.32±12.79	76.64±10.3	78.04±7.45	80.97±14.84	78.38±10.86
Painted-metal-sheets	96.36±3.3	87.29±14.74	99.78±0.38	99.87±0.28	99.56±0.87	99.43±1.09	99.69±0.78	99.77±0.61	**99.94±0.18**
Bare-Soil	69.24±8.96	80.3±12.31	88.27±11.34	87.61±11.12	87.58±7.76	85.54±8.06	84.24±8.9	94.69±5.09	**95.29±3.53**
Bitumen	74.06±8.22	88.13±7.81	83.41±11.37	84.48±11.6	80.89±11.25	80.3±13.56	**96.83±2.36**	93.28±4.6	93.44±3.63
Self-Blocking-Bricks	52.86±16.01	80.89±6.97	80.94±8.37	81.18±8.92	81.18±11.32	80.61±12.29	81.71±8.27	82.35±15.4	**86.25±12.11**
Shadows	89.64±5.47	78.14±6.69	**98.38±1.25**	98.22±1.57	98.03±1.19	98.1±1.7	93.32±3.32	97.79±1.43	97.94±3.17
OA(%)	71.36±6.47	73.74±2.83	78.16±4.59	78.73±4.41	79.69±6.16	81.22±4.45	82.73±4.42	82.84±6.8	**85.37±3.32**
AA(%)	73.02±4.14	75.0±2.67	83.06±2.78	82.95±2.55	83.1±3.43	84.04±2.05	86.1±2.34	87.56±2.91	**88.63±2.99**
Kappa(%)	70.38±6.57	70.62±4.37	80.88±4.51	80.75±4.68	80.2±4.12	81.62±3.54	82.62±4.98	87.5±4.74	**87.62±5.71**
mF1(%)	70.25±5.1	73.11±2.19	77.28±4.32	78.24±3.18	79.34±4.12	81.2±3.54	82.57±4.48	82.91±4.23	**85.08±3.66**

such as "Highway" and "Railway." As illustrated in Fig. 10, PIMamba yields clearer boundaries and more accurate regional recognition than Transformer-based and other Mamba variants, demonstrating its superior effectiveness on complex urban scenes.

F. Generality of Physical Constraints

To validate the generality of our proposed PIMamba framework, we further introduced physical constraints into traditional methods, including SVM, the CNN-based 3DCNN, and the Transformer-based ViT. We conducted comparative experiments on the IP dataset, and the results are

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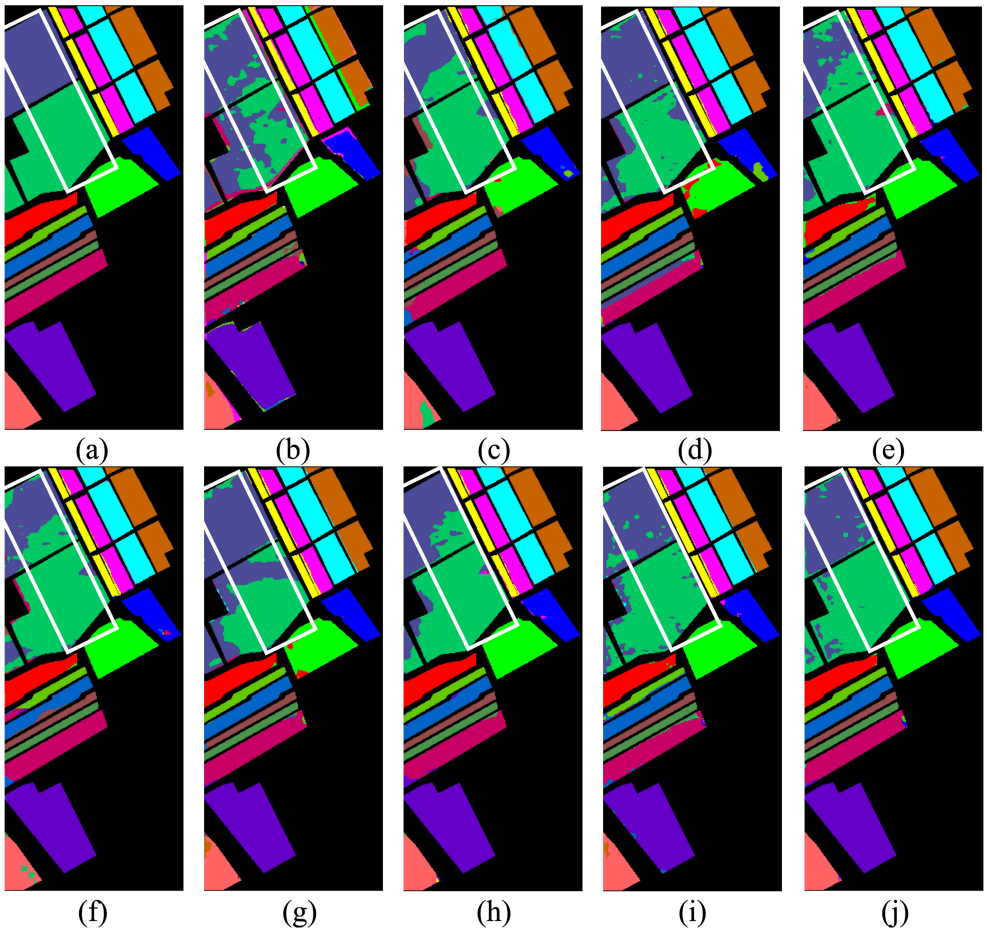


Fig. 9. Qualitative visualization of the classification map for the SA dataset. (a) Ground truth. (b) 3DCNN. (c) HybridSN. (d) SpectralFormer. (e) SSFTT. (f) GSC-ViT. (g) MambaHSI. (h) S^2Mamba. (i) GraphMamba. (j) PIMamba (ours).

TABLE III
QUANTITATIVE RESULT (ACC% ± STD%) OF SA DATASET (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	3DCNN	HybirdSN	SpectralFormer	SSFTT	GSC-ViT	MambaHSI	S^2Mamba	GraphMamba	PIMamba
Brocoli-green-weeds-1	85.43±8.66	97.09±4.85	97.66±3.26	97.08±3.7	97.34±4.11	97.38±3.81	95.55±6.09	96.82±3.11	**98.97±1.41**
Brocoli-green-weeds-2	95.34±3.09	93.93±7.43	98.56±3.84	98.4±4.62	99.05±2.46	97.81±3.41	99.68±0.74	98.65±3.34	**99.93±0.15**
Fallow	77.0±12.1	91.43±9.24	93.38±10.5	92.72±11.04	93.55±9.2	89.76±15.38	98.0±2.72	97.55±4.63	**98.84±2.2**
Fallow-rough-plow	94.3±6.38	99.53±1.09	98.75±2.03	98.77±2.18	98.82±1.99	99.22±1.15	99.43±0.69	98.52±1.48	**99.68±0.34**
Fallow-smooth	91.23±5.61	95.64±2.9	**97.29±1.81**	97.23±2.01	97.23±1.97	97.07±2.17	96.9±2.04	95.46±2.36	96.58±1.66
Stubble	99.88±0.21	97.12±4.07	99.58±0.57	99.55±0.62	99.58±0.52	99.73±0.34	99.81±0.35	99.92±0.08	**99.96±0.05**
Celery	85.84±6.56	96.39±7.99	98.68±1.4	98.67±1.33	98.55±1.5	98.96±0.98	98.78±1.37	99.0±1.09	**99.47±0.62**
Grapes-untrained	62.0±13.33	82.2±9.71	79.7±11.15	80.45±11.01	79.27±9.7	79.79±8.3	80.26±8.61	84.97±7.75	**87.47±8.23**
Soil-vineyard-develop	94.84±1.89	99.26±0.84	99.56±0.84	99.68±0.48	99.63±0.36	99.31±1.25	99.36±0.75	100.0±0.0	**100.0±0.0**
Corn-senesced-green-weeds	82.86±8.74	89.33±3.69	88.69±9.14	89.17±8.26	88.98±8.69	89.7±8.75	92.44±6.03	91.4±4.75	**93.02±5.31**
Lettuce-romaine-4wk	91.83±9.04	92.19±7.07	92.0±9.97	90.24±10.74	92.96±7.8	94.98±6.62	95.87±6.1	**98.48±1.94**	97.65±5.12
Lettuce-romaine-5wk	94.74±2.93	97.01±2.53	98.51±3.16	99.11±1.51	98.63± 3.06	99.81±0.31	99.61±0.7	99.72±0.81	**99.92±0.19**
Lettuce-romaine-6wk	98.08±1.53	96.16±5.53	99.19±1.6	99.2±1.62	99.16±1.65	99.53±0.73	99.72±0.5	**99.8±0.32**	99.72±0.33
Lettuce-romaine-7wk	96.64±2.46	96.45±3.92	98.96±0.93	99.1±0.63	98.9±0.94	98.82±1.98	99.39±0.43	98.58±1.37	**99.7±0.31**
Vinyard-untrained	68.4±9.91	70.73±21.03	83.85±15.08	83.16±15.1	83.71±12.5	87.71±9.8	87.84±9.42	88.07±15.9	**92.66±4.63**
Vinyard-vertical-trellis	86.19±6.12	94.96±7.98	97.6±1.78	97.8±1.53	97.36±2.12	97.06±2.27	96.34±2.5	**98.46±1.6**	97.76±2.3
OA(%)	82.01±1.78	89.54±2.93	91.82±1.46	91.02±1.48	91.76±1.35	92.31±1.29	92.95±1.97	93.98±1.54	**95.52±1.27**
AA(%)	87.66±1.49	93.09±1.52	95.12±1.56	95.02±1.54	95.17±1.25	95.42±1.22	96.19±0.76	96.59±0.82	**97.57±0.58**
Kappa(%)	87.27±2.05	92.27±2.57	94.07±2.05	94.0±2.07	93.93±1.75	94.87±2.0	96.4±1.82	95.93±1.38	**97.53±1.08**
mF1(%)	86.0±3.89	92.67±1.93	94.89±1.21	94.0±1.52	94.23±1.34	95.28±1.03	95.38±1.32	95.89±0.99	**97.22±0.75**

shown in Table V. It can be observed that after incorporating physical constraints, the OA of PI-SVM increased by 3.87% compared to the original SVM, PI-3DCNN improved by 3.06% compared to 3DCNN, and PI-ViT improved by

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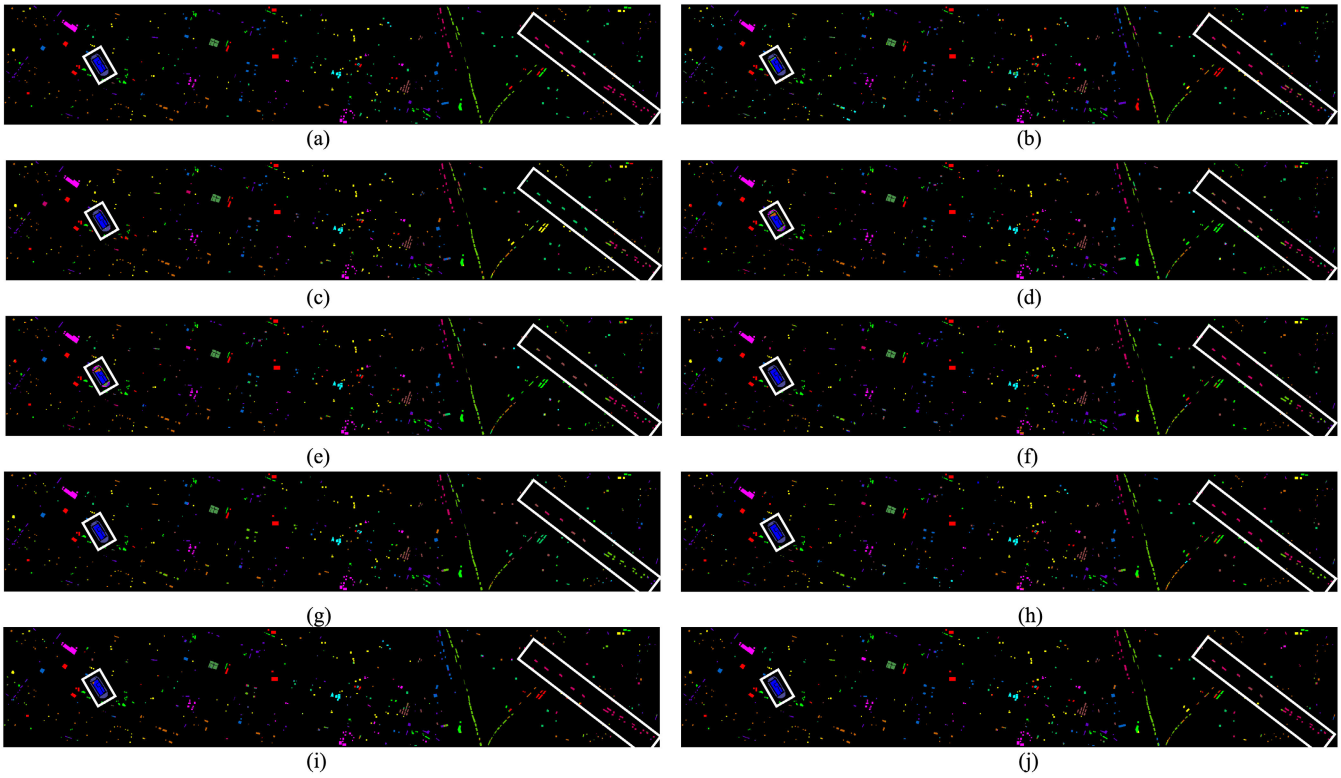


Fig. 10. Qualitative visualization of the classification map for the Houston dataset. (a) Ground truth. (b) 3DCNN. (c) HybridSN. (d) SpectralFormer. (e) SSFTT. (f) GSC-ViT. (g) MambaHSI. (h) S^{2}Mamba. (i) GraphMamba. (j) PIMamba (ours).

TABLE IV

QUANTITATIVE RESULT (ACC% ± STD%) OF HOUSTON DATASET (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	3DCNN	HybirdSN	SpectralFormer	SSFTT	GSC-ViT	MambaHSI	S^{2}Mamba	GraphMamba	PIMamba
Healthy-Grass	75.57±16.78	81.03±14.94	89.64±5.56	90.6±5.53	89.86±5.99	89.34±5.29	85.78±6.08	89.98±5.6	**90.8±4.69**
Stressed-Grass	74.53±7.9	78.04±11.48	83.42±6.94	82.49±8.2	82.32±8.77	86.85±8.06	87.26±7.8	83.0±8.39	**92.65±5.86**
Synthetic-Grass	94.85±3.67	90.37±15.19	92.26±11.99	92.3±11.82	92.26±12.54	**98.99±1.45**	91.06±11.52	97.26±2.15	97.61±2.29
Tree	68.14±11.73	89.82±7.85	88.17±10.2	88.1±10.2	85.23±8.36	90.01±6.48	91.06±5.59	90.52± 3.17	**92.1±5.05**
Soil	94.32±2.81	92.62±9.58	96.5±3.98	96.51±3.99	96.02±4.89	97.63±3.34	97.69±3.45	99.1±1.53	**99.5±0.77**
Water	57.32±9.98	75.94±10.3	82.26±3.27	82.29±3.15	81.03±6.38	86.42±7.5	86.84±5.77	85.16±6.39	**87.41±5.56**
Residential	63.17±20.36	71.6±9.65	71.76±10.47	71.84±10.58	72.2±1.54	71.43±13.27	73.48±12.41	74.1±15.18	**81.59±5.86**
Commercial	42.04±15.15	32.76±11.45	30.33±11.91	30.38±11.94	**46.38±15.94**	37.14±13.96	40.2±8.57	40.99±8.71	40.73±9.45
Road	64.17±8.85	70.92±11.75	71.08±11.47	70.22±11.38	71.55±13.06	76.11±10.7	77.13±8.11	70.16±10.73	**78.79±10.63**
Highway	56.74±14.25	54.08±15.75	61.16±11.78	61.25±11.73	58.74.25±13.42	56.34±10.65	60.82±9.01	64.54±11.29	**66.51±11.73**
Railway	52.8±10.53	59.07±9.17	58.07±7.05	58.41±6.75	59.82.41±12.6	67.68±10.03	68.92±10.35	72.22±7.43	**78.1±7.43**
Parking-Lot-1	59.47±15.04	52.9±11.29	59.21±15.03	59.01±15.67	62.88±9.01	58.24±9.42	59.87±13.86	57.24±11.55	**66.59±13.6**
Parking-Lot-2	81.61±9.79	83.37±5.74	86.52±9.24	86.43±9.33	88.13±7.79	**88.3±6.01**	83.68±9.91	87.89±4.65	87.35±6.13
Tennis-Court	91.23±7.02	97.34±4.16	99.25±1.74	99.27±1.67	96.69±2.75	99.54±1.23	99.88±0.16	**99.95±0.15**	99.59±0.66
Running-Track	73.19±20.84	83.21±16.48	80.87±18.3	81.12±24.34	78.57.12±16.25	94.65±5.3	89.6±18.85	97.84±2.83	**98.43±1.68**
OA(%)	67.92±2.77	71.43±3.84	73.9±3.76	73.78±3.74	74.12±3.85	76.74±2.95	76.98±2.96	77.78±2.8	**80.8±1.6**
AA(%)	69.94±2.93	74.2±3.05	76.7±3.27	76.62±3.27	76.83±3.3	79.91±2.34	79.55±2.69	80.66±2.21	**83.18±1.32**
Kappa(%)	66.0±4.03	71.14±4.68	74.5±3.75	74.5±3.75	74.64±3.49	78.57±3.93	77.93±5.04	76.21±4.81	**81.29±2.34**
mF1(%)	68.95±2.78	71.9±3.11	75.12±4.05	75.03±3.21	75.26+-3.75	77.88±3.61	77.93±3.68	78.33±2.59	**82.09±1.29**

3.26% compared to ViT. The three models also demonstrated significant improvements in AA and Kappa metrics. This indicates that physical constraints can effectively enhance the classification performance of existing methods and improve the generalization ability of models in small sample scenarios.

G. Ablation Study

To validate the contribution and synergistic effects of each module in the PIMamba framework, we conducted ablation studies on four datasets. The results are presented in Table VI. In this study, we sequentially removed the SpeLoss, SpaLoss,

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TABLE V
QUANTITATIVE RESULT (ACC% ± STD%) OF PHYSICAL-CONSTRAINED MODELS ON IP DATASET (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Metric	SVM	PI-SVM	3DCNN	PI-3DCNN	ViT	PI-ViT
OA(%)	55.6±4.44	**59.47±3.24**	59.06±3.44	**62.12±3.52**	67.65±4.37	**70.91±4.77**
AA(%)	66.64±3.21	**71.32±4.02**	73.28±2.13	**75.13±2.87**	79.92±2.26	**81.53±2.73**
Kappa(%)	65.2±3.71	**72.68±3.42**	71.6±2.97	**73.57±3.94**	78.47±2.09	**80.88±2.24**

TABLE VI
OA RESULT (ACC% ± STD%) OF ABLATION STUDY ON FOUR DATASETS (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Exp	SpeLoss	SpaLoss	DPA	Indian Pines	University of Pavia	Salinas Scene	Houston 2013
1	✓			75.33±2.59	83.3±3.73	92.61±0.78	77.36±2.81
2	✓		✓	76.29±3.84	83.71±2.18	93.45±1.75	78.28±2.24
3		✓		76.7±2.73	83.98±3.76	93.17±1.94	78.11±2.87
4		✓	✓	77.21±3.55	84.34±2.32	93.98±1.45	78.69±3.95
5	✓	✓		77.96±2.82	84.67±2.89	94.34±1.93	79.03±2.03
6	✓	✓	✓	79.79±2.59	85.37±3.32	95.52±1.27	80.8±1.6

and DPA mechanism to analyze their impact on classification performance.

Taking the IP dataset as an example, when only SpeLoss was employed (Exp 1), the OA reached 75.33%. SpeLoss effectively constrains spectral features with physical rules, capturing correlations between adjacent spectral bands and enhancing spectral modeling. When DPA was used to adjust the weight between SpeLoss and classification loss (Exp 2), OA increased by 0.96%.

With only SpaLoss applied (Exp 3), OA achieved 76.7%. SpaLoss constrains spatial features with physical rules, improving the model's capability to capture the spatial distribution of ground objects. When DPA adjusted the weight between SpaLoss and classification loss (Exp 4), OA increased by 0.51%.

When both SpeLoss and SpaLoss were simultaneously applied (Exp 5), OA further improved to 77.96%. Finally, with the introduction of the DPA mechanism (Exp 6), OA reached 79.79%. The DPA mechanism adaptively optimizes the weights of SpeLoss and SpaLoss, effectively balancing the learning of spectral and spatial features. This dynamic adjustment not only improves convergence efficiency during training but also enhances the model's robustness in few-shot scenarios.

H. Model Complexity Analysis

The model complexity analysis revealed significant differences in the number of parameters and computational efficiency among various methods. The results are shown in Table VII. While traditional 3DCNN demonstrates the lowest parameters (16.14k in IP) and FLOPs (0.006–0.012G), its shallow architecture limits feature extraction capability. HybridSN's 3-D–2-D hybrid design results in excessively high parameters (5124.3k, 15× SpectralFormer's 161.9–356.37k) although its fixed 0.496G FLOPs pattern indicates hardware optimization potential. Remarkably, state-space-based S^2Mamba and PIMamba achieve groundbreaking efficiency: on UP, PIMamba attains superior performance with 43.15k parameters (89.5% reduction versus MambaHSI) and 9.96G FLOPs (25.4% of MambaHSI); in large-scale Houston scenarios, its 46.3k parameters (88.9% less than MambaHSI) combined with 38.73G FLOPs deliver a 2.64× improvement in FLOPs/Params ratio. This co-optimization of parameters and computations provides an ideal solution for resource-constrained applications.

V. DISCUSSION

A. Advantages

The advantages of PIMamba extend beyond quantitative performance, highlighting its interpretability, portability, and practical value in few-shot hyperspectral classification.

1) *Physical Interpretability and Consistency:* PIMamba models hidden-state evolution through diffusion-inspired dynamics and introduces spectral–spatial physics-informed constraints into the learning process. This design aligns feature propagation with the intrinsic structure of hyperspectral data, improving interpretability and stability under few-shot conditions.
2) *Independence From External Pretraining:* Unlike many Transformer-based approaches, PIMamba is trained entirely within the target domain without relying on large-scale external datasets or pretrained models. This design improves portability and makes the model suitable for data-limited remote sensing scenarios.
3) *Lightweight and Efficient Architecture:* Benefiting from physics-guided regularization, PIMamba achieves strong representation with compact model complexity. The model contains only 43–56k parameters (about 10%–13% of MambaHSI) and reduces FLOPs by approximately 65%–75%, enabling efficient deployment on edge or airborne platforms.
4) *Cross-Model Generalizability:* The proposed physics-informed constraint framework is highly versatile. When applied to models such as SVM, 3DCNN, and ViT, the same spectral–spatial losses consistently improve classification performance, demonstrating their potential as a general regularization strategy.

B. Limitations

Although PIMamba demonstrates strong interpretability and competitive performance in few-shot HSI classification, several limitations remain.

First, the physical constraints adopted in this work rely on predefined diffusion-inspired priors that approximate the spectral–spatial continuity of hyperspectral data. While these priors provide effective regularization, they may not fully capture the complex physical interactions in real-world scenes. Future work may explore more adaptive or learnable physical operators to improve modeling flexibility.

Second, similar to many FSL methods, the model can still be sensitive to the selection of limited training samples. For example, on the IP dataset, the standard deviation of the "Corn-mintill" class reaches 13.13%, indicating variability under different sampling configurations. Future work could

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TABLE VII
COMPARATIVE ANALYSIS OF MODEL PARAMETERS AND FLOPS ACROSS DATASETS

Datasets	Metrics	3DCNN	HybirdSN	SpectralFormer	SSFTT	GSC-ViT	MambaHSI	S^2Mamba	GraphMamba	PIMamba
Indian Pines	Prarms(K)	16.14	5124.3	346.36	153.22	638.48	425.55	80.16	1242.45	56.43
	FLOPs(G)	0.012	0.496	7.58	1.59	0.043	4.49	6.45	3.28	1.98
University of Pavia	Prarms(K)	7.14	5124.3	161.9	152.77	152.71	412.24	78.48	1214.67	43.15
	FLOPs(G)	0.006	0.496	2.6	1.59	0.009	39.18	19.2	8.15	9.96
Salinas Scene	Prarms(K)	16.4	5124.3	356.37	153.22	179.02	426.07	81.35	1283.63	55.92
	FLOPs(G)	0.012	0.496	7.84	1.59	0.013	23.86	25.8	7.41	8.46
Houston 2013	Prarms(K)	11.99	5124.3	226.29	153.16	163.6	418.26	79.47	1237.58	46.3
	FLOPs(G)	0.009	0.496	4.41	1.59	0.011	132.57	127.4	38.6	38.73

investigate more robust sampling strategies or uncertainty-aware training mechanisms to further enhance stability under extremely limited supervision.

VI. CONCLUSION

In this article, we propose PIMamba, a physics-informed HSI classification framework designed for FSL scenarios. The proposed method integrates diffusion-inspired physical modeling into the state-space-based Mamba architecture, guiding hidden-state evolution with spectral–spatial dynamics. A dual-branch structure jointly models spectral and spatial dependencies, while SpeLoss and SpaLoss enforce physically consistent feature representations. In addition, a DPA mechanism adaptively balances the contributions of spectral and spatial constraints during training.

Extensive experiments on four benchmark datasets (IP, UP, SA, and Houston) demonstrate that PIMamba achieves SOTA performance under few-shot settings without relying on external annotated data or large-scale pretrained models. The proposed physics-informed constraint framework also shows strong generalizability when applied to other models such as SVM, 3DCNN, and ViT.

Future work may explore adaptive physical operators, multimodal data integration, and extensions of the proposed framework to more complex remote sensing tasks such as change detection and object detection.

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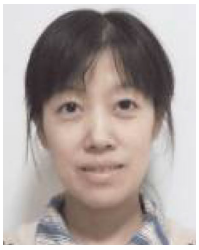
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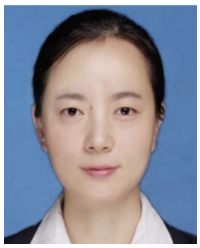
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